

MATHEMATICS PROBLEM WITH ANSWERS AND SOLUTIONS

INSTRUCTION: Select the correct answer for each of the following questions. Mark only one answer for each item by shading the box corresponding to the letter of your choice on the answer sheet provided. STRICTLY NO ERASURES ALLOWED. Use pencil No. 1 only.

1. How do call an energy required to move 1 Coulomb of charge through an element.
a) Current **b) Voltage** c) Power d) Resonance

Answer: b) Voltage $V = \frac{dw}{dq}$ where q = charge in C w = energy in Joules

2. This is a number sequence where the succeeding term is obtained by adding the last pair of preceding terms such as the sequence (1, 1, 2, ,3 5, 8 ...). How do you call this number sequence?
a) Euler's number b) Fermat number **c) Fibonacci number** d) Fourier series

Answer: c) Fibonacci number

3. Determine the sum of the coefficient of the variables in the expression $(2x + 3y - z)^8$.
a) 65 536 b) 56 563 c) 63 655 d) 66 535

Solution: sum = $[2(1) + 3(1) - 1]^8 = 56\ 563$

4. If the roots of an equation are zero, then, how do you classify the solutions?
a. Extraneous solutions **b. Trivial solutions** c. Conditional solutions d. Ambiguous solutions

Answer: b) Trivial Solutions

5. Find the voltage drop to move 2 C of charge from point a to point b that requires – 30 Joules of energy.
a) – 15 V b) + 15 V c) – 5 V d) + 5 V

Solution: $v = \frac{\Delta w}{\Delta q} = \frac{-30\text{ J}}{2\text{ C}} = -15\text{ V}$

6. This is the process of expressing a polynomial as the product of another polynomial or monomial of lower degree. What is this mathematical process?
a) Decomposition b) Rationalization **c) Factoring** d) Polynomial damping

Answer: c) Factoring

7. This is a point where the concavity of a curve changes or when the slope of the curve is neither increasing nor decreasing. What is this point commonly called?
a) Maximum point b) Minimum point c) Point of tangency **d. Point of inflection**

Answer: d) point of inflection

8. How do you call the axis of the hyperbola that passes through the center, the foci and vertices?
a) Transverse axis b) Conjugate Axis c) Asymptotic axis d) Major Axis

Answer: a) Transverse axis

9. How many permutations can be made out of the letters in the word DIEGO taken 3 at a time?
a) 50 ways **b) 60 ways** c) 120 ways d) 80 ways

Solution: $P(n, r) = P(5, 3) = \frac{n!}{(n - r)!} = \frac{5!}{(5 - 3)!} = 60\text{ ways}$

10. What is the temperature in degrees Celsius of a molecule at absolute zero?
a) 0 °C **b) – 273 °C** c) – 32 °C d) 273 °C

Answer: b) – 273 °C

11. If at an angle measures x degrees, what is the measure in radians?

- a) $\frac{\pi}{180}x$ b) $\frac{180}{\pi}x$ c) $\frac{180\pi}{x}$ d) $\frac{180}{\pi x}$

Solution: a) $\frac{x\pi}{180^\circ}$

12. In electricity, how do you call the rate of charge flow?

- a) Potential difference **b) Current** c) Voltage d) Power

Answer: b) Current $i = \frac{dq}{dt}$ where q = charge, C; t = time, sec.

13. What is a number, which could not be expressed as a quotient of two integers?

- a. Natural b. Rational **c. Irrational** d. Surd

Answer: c) Irrational

14. This law states, “The voltage, v, across a resistor is directly proportional to the current, i, flowing through the resistor”. How do you call this law?

- a) Kirchhoff’s Law **b) Ohm’s Law** c) Ampere’s Law d) Gauss’ Law

Answer: b) Ohm’s Law

15. How do you call an angle that is greater than 180 degree but less than 360 degrees?

- a) Complex **b) Reflex** c) Obtuse d) Exterior

16. This law in electrical circuits states, “The algebraic sum of currents entering a node (or a closed boundary) is zero”. How do you call this law?

- a) Kirchhoff’s current law** b) Ohm’s current law
c) Kirchhoff’s voltage law d) Ohm’s voltage law

Answer: a) Kirchhof’s current law

17. How do you call the opposite of the prefix nano?

- a) Peta b) Tera **c) Giga** d) Hexa

Solution: c) Giga Note: The prefix nano represents 10^{-9} which is the opposite of giga (10^9).

18. What do you call a triangle having three unequal sides?

- a) Obtuse b) Oblique **c) Scalene** d) Isosceles

Answer: c) Scalene

19. An area of 25.6 cm² on the map represents a lot having an area of 640 m². If the scale on the map is 1:x, what is the value of x?

- a) 50 b) 100 **c) 500** d) 900

Solution: $\frac{\text{Map Area}}{\text{Actual Area}} = \left(\frac{1}{x}\right)^2 = \frac{25.6}{640(100)^2}$ x = 500

20. If the $z = \sqrt{1 - \sqrt{1 - \sqrt{1 - \dots}}}$, what is the value of z?

- a) 0.453 **b) 0.618** c) 0.816 d) 0.681

Solution: Squaring both sides of the equation

$$\left(z = \sqrt{1 - \sqrt{1 - \sqrt{1 - \dots}}}\right)^2$$

$$z^2 = \left(1 - \sqrt{1 - \sqrt{1 - \sqrt{1 - \dots}}}\right)^2 \quad \text{but } \sqrt{1 - \sqrt{1 - \sqrt{1 - \dots}}} = z$$

$$z^2 = 1 - z$$

$$z^2 + z - 1 = 0$$

$$z = \frac{-1 \pm \sqrt{(1)^2 - 4(1)(-1)}}{2(1)} = \begin{cases} 0.618 \\ -1.619 \end{cases}$$

21. Solve for the value of x from the following equation: $x^{x^{x^{x^{\dots}}}} = 10$.
- a) **1.258925** b) 1.892525 c) 1.85925 d) 1.528925

Solution: $\ln x^{x^{x^{x^{\dots}}}} = \ln 10 \rightarrow x^{x^{x^{x^{\dots}}}} \ln x = \ln 10 \rightarrow \text{but } x^{x^{x^{x^{\dots}}}} = 10$

$$10 \ln x = \ln 10 \rightarrow \ln x^{10} = \ln 10 \rightarrow x^{10} = 10 \quad x = (10)^{\frac{1}{10}} = 1.258925$$

22. This law in electrical circuits state, “The algebraic sum of all voltages around a closed path (or loop) is zero”. How do you call this law?
- a) Kirrchoff’s current law b) Ohm’s current law
c) **Kirchhoff’s voltage law** d) Ohm’s voltage law

Answer: Kirchhoff’s voltage law

23. How do you call the distance of a point from the y-axis?
- a) Polar distance b) Coordinate c) **Abscissa** d) Ordinate
24. In electricity, it is an ability of an element to resist the flow of current. What is this?
- a) Resonance b) Conductance c) Inductance d) **Resistance**

Answer: d) Resistance

25. In electrical circuits, how do you call the reciprocal of resistance?
- a) Resonance b) **Conductance** c) Inductance d) **Resistance**

Answer: b) Conductance

26. In electrical, what is the SI unit of conductance?
- a) Ohm b) Mho c) **Siemens** d) Ampere

Answer: c) Siemens 1 Mho = 1 Siemens

27. If $4y = 3x$, then, what is the value of $\frac{3x^2}{4y^2}$?

a) $\frac{4}{3}$ b) $\frac{3}{4}$ c) $\frac{2}{3}$ d) $\frac{3}{2}$

Solution: $\frac{x}{y} = \frac{4}{3} \rightarrow \left(\frac{x}{y} = \frac{4}{3}\right)^2 \rightarrow \frac{x^2}{y^2} = \frac{16}{9}$ Then, $\frac{3x^2}{4y^2} = \frac{3}{4} \left(\frac{16}{9}\right) = \frac{4}{3}$

28. Which of the following is the equivalent of 1 Ampere?
- a) **1 Coulomb per second** b) 1 Joule per Coulomb c) 1 Volt per Ampere d) 1 Ampere per Coulomb

Answer: a) 1 Coulomb per second

$$\begin{array}{ll} 1 \text{ Ohm} = 1 \text{ Volt/Ampere} & 1 \text{ Siemens} = 1 \text{ Ampere/Volt} \\ 1 \text{ Volt} = 1 \text{ Joule/Coulomb} & 1 \text{ Ampere} = 1 \text{ Coulomb per second} \end{array}$$

29. This is the measure of central tendency defined as the most frequent score. How do you call this measure of central tendency?

- a) Median **b) Mode** c) Mean d) Deviation

30. When the expression $x^4 + ax^3 + 5x^2 + bx + 6$ is divided by $(x - 2)$, the remainder is 16. When it is divided by $(x + 1)$ the remainder is 10. Determine the value of the constant a.
a) 6 b) 9 c) 5 **d) 7**

Solution: $f(x) = x^4 + ax^3 + 5x^2 + bx + 6$

$$f(2) = (2)^4 + a(2)^3 + 5(2)^2 + b(2) + 6 = 16 \qquad 8a + 2b = -26 \quad \rightarrow \quad 4a + b = -13 \qquad \text{Eq. 1}$$

$$f(-1) = (-1)^4 + a(-1)^3 + 5(-1)^2 + b(-1) + 6 = 10$$

$$-a - b = -2 \quad \rightarrow \quad a + b = 2 \quad \rightarrow \quad b = 2 - a \qquad \text{Eq. 2}$$

Substituting 2 to 1, $4a + 2 - a = -13 \rightarrow 3a = -15$

$$a = \frac{-15}{3} = -5 \qquad \text{the, } b = 2 - (-5) = 7$$

31. What is the logarithm of a negative number?
a) Complex number b) Real number
c) Irrational number d) Imaginary number

Solution: The logarithm of a negative number can be evaluated as follows:

$$\log(-x) = \log(x)(-x) \quad \text{where, } i^2 = -1$$

$$\log(-x) = \log(x)(i)^2 = \log x + \log i^2 = \log x + 2 \log i$$

Considering the exponential form of an imaginary number, $i = e^{i\left(\frac{\pi}{2}\right)}$

$$\log(-x) = \log x + 2 \log e^{i\left(\frac{\pi}{2}\right)} = \log x + 2i\left(\frac{\pi}{2}\right) \log e = \log x + \pi i \log e$$

Thus, the logarithm of a negative number is a complex number.

32. Which of the following is the equivalent of 1 mil?
a) One-tenth of an inch **b) One-thousandth of an inch**
c) One millionth of an inch c) One-half of an inch

Answer: a) One-thousandth of an inch

33. This is the case of a solution of a plane triangle where the given data leads to two solutions. How do you call this case?
a) Ambiguous case b) Quadratic case c) Extraneous case d) Conditional case

Answer: a) Ambiguous case

34. What is the value of k to make the expression $kx^2 - 3kx + 9$ a perfect square?
a) 2 **b) 4** c) 3 d) 5

Solution: The expression $Ax^2 + Bx + C$ is a perfect square if $B^2 = 4AC$.

$$\text{From the given expression: } A = k \qquad B = -3k \qquad C = 9$$

$$\text{Then, } (-3k)^2 = 4(k)(9) \quad \rightarrow \quad k = 4$$

35. It is a type of polygon in which each interior angle must be less than or equal to 180° , and all vertices 'point outwards' away from the interior. How do you call this polygon?
a) Concave Polygon **b) Convex polygon**
c) Regular polygon d) Irregular polygon

Answer: b) Convex polygon

36. Two times the mother's age is 8 more than six times her daughter's age. Ten years ago, the sum of their ages was 44. What is the daughter's age?

a) 15 yrs old b) 18 yrs old c) 12 yrs old d) 16 yrs old

Solution: Let x = daughter's age y = mother's age

$$\text{Present Ages: } 2y = 6x + 8 \quad \rightarrow \quad y = 3x + 4 \quad \text{eq. 1}$$

$$\begin{aligned} \text{Ten years ago: } (x-10) + (y-10) &= 44 \\ x + y &= 64 \end{aligned} \quad \text{eq. 2}$$

$$\text{Substituting 1 to 2, } x + 3x + 4 = 64 \quad \rightarrow \quad 4x = 60 \quad \rightarrow \quad x = \frac{60}{4} = 15$$

37. How do you call a line segment from the center of a regular polygon to the midpoint of a side? It is also called as inradius, the radius of an incircle.

a) Radius **b) Apothem** c) Circumradius d) Inradius

Answer: b) Apothem

38. If the daily wage of CPM and a Plumber are in the ratio 2:1. In a day, a CPM has to work 8 hrs but the Plumber only 6 hrs. Determine the ratio of their hour wages.

a. 4:3 b. 5:3 **c. 3:2** d. 8:3

Solution: Let x = hourly wage of CPM y = hourly rate of the Plumber

$$\frac{x}{y} = \frac{\frac{x}{8}(8)}{\frac{y}{8}(6)} = \frac{x}{\frac{6}{8}y} = \frac{8x}{6y} = \frac{2}{1}; \quad \frac{x}{y} = \frac{6}{8} \left(\frac{2}{1} \right) = \frac{6}{4} = \frac{3}{2}$$

Then, the ratio is 3:2

39. Which of the following is the equivalent of one circular mil?

a) $\frac{\pi}{4}(0.01)^2$ **b) $\frac{\pi}{4}(0.001)^2$** c) $\frac{\pi}{4}(0.0001)^2$ d) $\frac{\pi}{4}(0.00001)^2$

Answer: b) $\frac{\pi}{4}(0.001)^2$

A circular mil is the area of a circle whose diameter is 1 mil or 0.001 inch.

40. How do you call a method of describing a set where the elements are separated by commas and enclosed by braces?

a) Tabular or roster form b) Equal form
c) Rule form d) Equivalent form

Answer: a) Tabular or roster form

Two Ways of Describing a Set: a) Tabular or roster form; and b) Rule form, a method which makes use of the description $\{x \mid \dots\}$.

41. Which of the following is the equivalent of the expression $2\text{Log}_b 6 - \text{Log}_b 4$?

a) $2\text{Log}_b 3$ b. $3\text{Log}_b 2$ c. $2\text{Log}_b 2$ d. $3\text{Log}_b 3$

$$\text{Solution: } 2\log_b 6 - \log_b 4 = \log_b 36 - \log_b 4 = \log_b \left(\frac{36}{4} \right) = \log_b 9 = 2\log 3$$

42. It is a series of equal payments occurring at equal intervals of time where the first payment is made after several periods, after the beginning of the payment. How do you call this payment?
- a) **Deferred annuity** b) Delayed annuity
c) Progressive annuity d) Simple annuity

Answer: a) Deferred annuity

43. A 100-kg salt solution originally 4 % by weight salt in water is boiled to reduce water content until the concentration is 5 % by weight salt. How much water is evaporated?
- a) 20 b) 25 c) 15 d) 22.5

Solution:

100 kg	x	100 - x
4 % salt 96 % H ₂ O	- 100 % H ₂ O	= 5 % salt 95 % H ₂ O

In Salt:

$$4 \% (100) - 0 = 5 \% (100 - x) \rightarrow 400 - 500 - 5x \rightarrow \text{then, } x = 20 \text{ kg}$$

In Water (Checking):

$$96\% (100) - 100\% (x) = 95\% (100 - x) \rightarrow 1920 - 20x = 1900 - 19x \rightarrow \text{then, } x = 20 \text{ kg}$$

44. Tukmol can paint a fence of 50 % faster than Kikoy and 20 % faster than Tiburcio and together they can paint a given fence in 4 hours. How long will it take Tukmol to paint the same fence if he had to work alone?
- a) 11 b) 8 c) 9 **d) 10**

Solution:

Let, x = no. of hours for Tukmol alone to paint the fence
 y = no. of hours for Kikay to paint the fence
 z = no. of hours for Veronica

With the three of them finishing the work together in 4 hours,

$$\frac{1}{x}(4) + \frac{1}{y}(4) + \frac{1}{z}(4) = 1 \quad \rightarrow \quad \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{4} \quad \text{Eq. 1}$$

With Tukmol being 50 % than Kikay,

$$\frac{1}{x} + \frac{1}{y} + (0.50)\left(\frac{1}{y}\right) = \frac{2}{3y} \quad \rightarrow \quad \frac{1}{y} = \frac{2}{3x} \quad \text{Eq. 2}$$

With Tukmol being 20 % faster than Veronica,

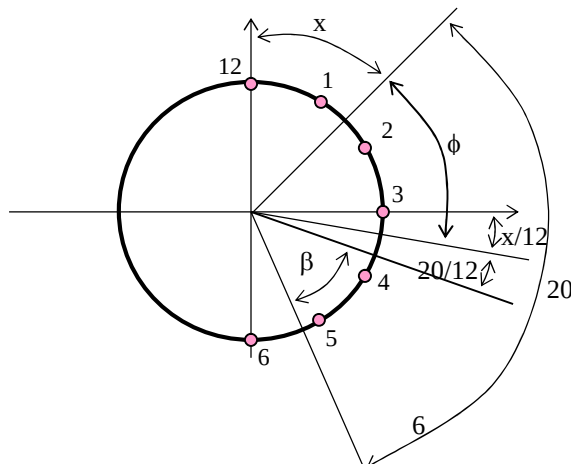
$$\frac{1}{x} = \frac{1}{z} + (0.20)\left(\frac{1}{z}\right) = \frac{6}{5x} \quad \rightarrow \quad \frac{1}{x} = \frac{5}{6x} \quad \text{Eq. 3}$$

Substituting $1/y$ in Eq. 2 and $1/z$ in Eq. 3 to Eq. 1,

$$\frac{1}{x} + \frac{2}{3x} + \frac{5}{6x} = \frac{1}{4} \quad \rightarrow \quad \frac{5}{2x} = \frac{1}{4} \quad \rightarrow x = 10 \text{ hours}$$

45. It is now between 3 and 4 o'clock and twenty minutes the minute hand will be as much as the hour-hand as it is now behind it. What is the time now?
- a) 3:06.36 b) 3:03.66 c) 3:36.06 d) 3:30.66

Solution:



$$i = \left(\frac{\frac{F}{P} - 1}{n} \right) (100\%) = \left(\frac{F - P}{nP} \right) 100\% = \left[\frac{45,937.5 - 37,500.00}{(1.5)(37,500.00)} \right] (100\%) = 15\%$$

51. What do you think is the negotiation of wage rates, conditions of employment, etc. by representatives of the labor force and management?

- a) Union trade Cooperative b) Union rally **c) Collective bargaining** d)

52. What is the simple interest rate if an investment of P37,500.00 accumulates to P45,937.5 in 18 months?

- a) 0.15** b) 0.2 c) 0.21 d) 0.3

Solution: $F = P(1 + in)$

$$i = \left(\frac{\frac{F}{P} - 1}{n} \right) (100\%) = \left(\frac{F - P}{nP} \right) 100\% = \left[\frac{45,937.5 - 37,500.00}{(1.5)(37,500.00)} \right] (100\%) = 15\%$$

53. What is the value of $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^3 - 1}$?

- a) 0** b) 0.25 c) 1.25 d) indeterminate

Solution: $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^3 - 1} = \lim_{x \rightarrow \infty} \frac{(x+1)(x-1)}{(x-1)(x^2 + x + 1)} = \lim_{x \rightarrow \infty} \frac{x+1}{x^2 + x + 1} = \lim_{x \rightarrow \infty} \frac{1}{\frac{x^2 + x + 1}{x+1}}$

$$\lim_{x \rightarrow \infty} \frac{1}{x + \frac{1}{x+1}} = \frac{1}{\infty + \frac{1}{\infty + 1}} = \frac{1}{\infty} = 0$$

54. What is the present value of P5,000.00 due in 8 years if money is worth 12% compounded semi-annually?

- a) P1,896.23 b) P1,869.23 c) P1,698.23 **d) P1,968.23**

Solution: $P = \frac{F}{\left(1 + \frac{i_n}{m}\right)^{mn}} = \frac{5,000.00}{\left(1 + \frac{0.12}{2}\right)^{2(8)}} = P1,968.23$

55. This is a type of bond whose guaranty is in lieu on railroad equipment. What is this type of bond?

- a) Equipment b) Debenture bond c) Registered bond **d) Infrastructure bond**

Answer: d) Infrastructure bond

56. How do you call a type of bond where the corporation's owner name is recorded and the interest is paid periodically to the owners with their asking for it?

- a) Registered bond** b) Preferred bond c) Incorporator's bond d) Bail bond

Answer: a) Registered bond

57. Twice the middle digit of a three-digit number is the sum of the other two. If the number is divided by the sum of its digit, the answer is 56 and the remainder is 12. If the digits are reversed, the number becomes smaller by 594. Find the number.

- a) 852** b) 285 c) 825 d) 582

Solution: Let x, y, and z be the hundred's, ten's, and unit's digits, respectively.

The number is: $100x + 10y + z$

The reversed number is: $100z + 10y + x$

$$2y = x + z \quad \text{Eq. 1}$$

$$\frac{100x + 10y + z}{x + y + z} = 56 + \frac{12}{x + y + z} \rightarrow \frac{(100x + 10y + z) - 12}{x + y + z} = 56$$

$$100x + 10y + z - 12 = 56x + 56y + 56z$$

$$44x - 99y - 55z = 12 \quad \text{Eq. 2}$$

$$100z + 10y + x = 100x + 10y + z - 594$$

$$99z - 99x = -594$$

$$x = 6 + z \quad \text{Eq. 3}$$

Substituting x in Eq. 3 to Eq. 1,

$$2y = (x + z) + z = 6 + 2z$$

$$y = 3 + z \quad \text{Eq. 4}$$

Substituting x in Eq. 3 to Eq. 2,

$$44(6 + z) - 46(3 + z) - 55z = 12$$

$$\text{Then, } z = 2 \quad x = 8 \quad y = 5$$

The number is 852

Note: This problem can be solved by inspection.

58. A man left Sta. Rosa City to drive to Lopez, Quezon at 6:15 pm and arrived at 11:45 pm. If he averaged 50 kph and stopped 1 hour for dinner, how far is Lopez, quezon from Sta. Rosa City?
a) 225 km b) 522 km **c) 252 km** d) 215 km

Solution: Total time, 6:15 pm to 11:45 pm, = 5.5 hours

$$\text{Time of travel} = 5.5 - 1 = 4.5 \text{ hours}$$

$$\text{Distance, } S = Vt = 50 (4.5) = 225 \text{ km}$$

59. If x varies directly as y and inversely as z, and x = 14 when 7 = 7 and z = 2, find the value of x when z = 4 and y = 16.
a) 12 b) 18 c) 14 **d) 16**

$$\text{Solution: } x \propto \frac{y}{z} \rightarrow x = k \left(\frac{y}{z} \right) \quad k = x \left(\frac{z}{y} \right) = 14 \left(\frac{2}{7} \right) = 4$$

$$\text{Then, } x = k \left(\frac{y}{z} \right) = 4 \left(\frac{16}{4} \right) = 16$$

60. Lucy sold 100-pirated DVD. Eighty of them were sold at a profit of 30 % while the rest were sold at a loss of 40 %. What is the percentage gain or loss on the whole stock?
a) 15 % b) 20 % **c) 16 %** d) 12 %

Solution: Let x = the buying price of each DVD
Total Capital = 100x
Income for 80 DVD = 80 (x + 0.30x) = 104x
Income for 20 DVD = 20(x - 0.40x) = 12x

$$\text{Total sales} = 104x + 12x = 116x$$

67. The sum of the two interior angles of the triangle is equal to the third angle and the difference of the two angles is equal to $\frac{2}{3}$ of the 3rd angle. Find the 3rd angle.
 a) 60° b) 30° **c) 90°** d) 40°
- Solution: Let $x = 1^{\text{st}}$ angle $y = 2^{\text{nd}}$ angle $z = 3^{\text{rd}}$ angle
- $$x + y + z = 180 \quad \text{Eq. 1} \quad \quad \quad x + y = z \quad \text{Eq. 2}$$
- $$x - y = \left(\frac{z}{3}\right)z \quad \text{Eq. 3}$$
- Substituting Eq. 2 to Eq. 1, $z = 90^\circ$
68. Evaluate the integral of $(3x^2 + 9y^2)dxdy$ if the interior limit has an upper limit of y and a lower limit of 0, and whose outer limit has an upper limit of 2 and lower limit of 0.
 a) 50 b) 30 c) 45 **d) 40**
- Answer: d) 40
69. How do you call the integral of any quotient whose numerator is the differential of the denominator?
 a) Co-logarithm **b) Logarithm** c) Product d) Derivative
- Answer: b) Logarithm
70. If $a = b$, then $b = a$. This illustrates of which of the following axioms in algebra?
 a) Transitive axiom b) Reflexive axiom c) Replacement axiom **d) Symmetric axiom**
- Answer: d) Symmetric axiom
71. What is a regular polygon that has 27 diagonals?
a) Nonagon b) hexagon c) Pentagon d) Heptagon
- Answer: a) Nonagon
72. How do you call the formula used to compute the value of n factorial, which is in symbolic form $(n!)$, where n is large number?
 a) Matheson formula b) Diophantine formula c) Richardson-Duchman formula **d) Stirling's Approximation**
73. What is the reason why an ivory soap floats in water?
 a) All matter has mass
 b) The specific gravity of ivory soap is greater than that of water
 c) The density of ivory soap is unity
d) The specific gravity of ivory soap is less than that of water
74. If the product of the slopes of any two straight lines is negative 1, one of these lines is said to be:
 a) Parallel **b) perpendicular** c) Skew d) Non-intersecting
75. When two planes intersect with each other, the amount of divergence between the two planes is expressed by measuring the:
 a) Reflex angle **b) Dihedral angle** c) Polyhedral angle d) Plane angle
76. These are product or services that are required to support human life and activities, that will be purchased in somewhat the same quantity even though the price varies considerably.
 a) Utilities **b) Necessities** c) Luxuries d) Producer goods and services
77. The angle which the line of sight to the object makes with the horizontal which is above the eye of the observer is called as:
a) Angle of elevation b) Acute angle c) Angle of depression d) Bearing
78. This is a number sequence where the succeeding term is obtained by adding the last pair of preceding terms such as the sequence (1, 1, 2, 3, 5, 8 ...). How do you call this number sequence?
 a. Euler's number b. Fermat number **c. Fibonacci number** d. Fourier series

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a. Angles b. Diagonals **c. Sides** d. Number of nodes
99. When a line $y = mx + b$ slopes downwards from left to right, the slope m is:
a) Less than zero b) Greater than zero c) Equal to zero d) Equal to 1
100. The ratio between the quick assets to the current liabilities
a. Acid test ratio b. Current ratio c. Working amount d. Marginal ratio
101. It is the additional cost of producing one more unit of a product.
a. Sunk cost **b. Marginal cost** c. Variable cost d. Fixed cost
102. What is the quantity in addition to the economic order quantity use to anticipate sudden change of demand and the delay of the ordered quantity?
a. safety stock b. safety allowance c. tolerance stock d. none of these
103. The ratio of a number is 4:7 where number 4 is referred to as the antecedent. How do you call number 7 in this ratio?
a. Consequent b. Minuend c. Augend d. Dividend
104. What is a pair of angles when added together will be equal to 360° ?
a. Supplementary angle **b. Conjugate angle** c. Complimentary angle d. Explimentary angle
105. What is a line segment joining a vertex of a triangle to a point on the opposite side?
a. Cevian b. Secant c. Euclidian line d) Newton's line
106. How do you call a cylinder with elliptical cross section?
a. Ellipsoid **b. Cylindroid** c. Hyperboloid d. Paraboloid
107. For a given function, it is found that $f(t) = f(-t)$. What type of symmetry does $f(t)$ have?
a) Odd Symmetry **b) Even Symmetry** c) Rotational Symmetry d) Quarter-wave Symmetry
108. Which is true regarding the signs of the natural functions for angles between 90° and 180° ?
a) The tangent is positive b) The cotangent is positive c) The cosine is negative **d) The sine is negative**
109. What is the inverse natural function of the cosecant?
a) Secant **b) Sine** c) Cosine d) Tangent
110. What type of curve is generated by a point, which moves in uniform circular motion about an axis, while traveling with a constant speed, V , parallel to the axis?
a) A cycloid b) An epicycloids c) A hypocycloid **d) A helix**
111. Forty liters of a 60 % salt solution are produced to a 45 % solution. How much solution must be drained off and replaced with distilled water so that the resulting solution contains only 45 % solution.
a) 6 liters b) 8 liters **c) 10 liters** d) 12 liters

Answer: c) 10 liters

Solution: Let x represents the number of liters of salt solution to be drained off and replaced with distilled water.

60 % salt 40 % water	-	60 % salt 40 % water	+	0 % salt 100 % water	=	55 % salt 45 % water
40 liters		x liters		x liters		40 liters

Based on salt solution: $0.60(40) - 0.60x + (0.0)x = 0.45(40)$ $x = 10$ liters

Based on water solution: $0.40(40) - 0.40x + x = 0.55(40)$ $x = 10$ liters

112. How do you call the area bounded by two concentric circles?

- a) Disk **b) Annulus** c) Ring d) Sector

113. Determine, which of the following is incorrect between lease and outright purchase of equipment?

- a) Leasing fees needed working capital b) Leasing reduces maintenance and administrative expenses
c) **Leasing offers certain tax advantage** d) Leasing offers less flexibility with respect to technical obsolescence.

114. What do you call the replacement of the original cost of an investment?

- a) Pay off** b) Return on investment c) Breakeven d) Capital recovery

115. What do you think is the output or sales at which income is insufficient to equal operating cost?

- a) Break even point** b) Depreciation c) Investment d) Cash flow

116. What do you think is the lessening of the value of an asset due to a decrease in the quantity available as a coal, oil and timber in forests?

- a) Depletion** b) Amortization c) Depreciation d) Investment

117. How do you call the integral of any quotient whose numerator is the differential of the denominator?

- a) Co-logarithm **b) Logarithm** c) Product d) Derivative

118. If $a = b$, then $b = a$. This illustrates of which of the following axioms in algebra?

- a) Transitive axiom b) Reflexive axiom c) Replacement axiom **d) Symmetric axiom**

119. What is a regular polygon that has 27 diagonals?

- a) Nonagon** b) hexagon c) Pentagon d) Heptagon

120. How do you call the formula used to compute the value of n factorial, which is in symbolic form $(n!)$, where n is large number?

- a) Matheson formula b) Diophantine formula c) Richardson-Duchman formula **d) Stirling's Approximation**

121. What is the reason why an ivory soap floats in water?

- a) All matter has mass b) The specific gravity of ivory soap is greater than that of water
c) The density of ivory soap is unity **d) The specific gravity of ivory soap is less than that of water**

122. If the product of the slopes of any two straight lines is negative 1, one of these lines is said to be:

- a) Parallel **b) perpendicular** c) Skew d) Non-intersecting

123. When two planes intersect with each other, the amount of divergence between the two planes is expressed by measuring the:

- a) Reflex angle **b) Dihedral angle** c) Polyhedral angle d) Plane angle

124. These are product or services that are required to support human life and activities, that will be purchased in somewhat the same quantity even though the price varies considerably.

- a) Utilities **b) Necessities** c) Luxuries d) Producer goods and services

125. The angle which the line of sight to the object makes with the horizontal which is above the eye of the observer is called as:

- a) Angle of elevation** b) Acute angle c) Angle of depression d) Bearing

126. This is a number sequence where the succeeding term is obtained by adding the last pair of preceding terms such as the sequence (1, 1, 2, 3, 5, 8 ...). How do you call this number sequence?

- a. Euler's number b. Fermat number **c. Fibonacci number** d. Fourier series

127. If the roots of an equation are zero, then, how do you classify the solutions?

- a. Extraneous solutions **b. Trivial solutions** c. Conditional solutions d. Ambiguous solutions
128. This is the process of expressing a polynomial as the product of another polynomial or monomial of lower degree. What is this mathematical process?
a. Decomposition b. Rationalization **c. Factoring** d. Polynomial damping
129. This is a point where the concavity of a curve changes or when the slope of the curve is neither increasing nor decreasing. What is this point commonly called?
a. Maximum point b. Minimum point c. Point of tangency **d. Point of inflection**
130. How do you call the axis of the hyperbola that passes through the center, the foci and vertices?
a. Transverse axis b. Conjugate Axis c. Asymptotic axis d. Major Axis
131. What is a number, which could not be expressed as a quotient of two integers?
a. Natural b. Rational **c. Irrational** d. Surd
132. How do you call an angle that is greater than 180 degree but less than 360 degrees?
a. Complex **b. Reflex** c. Obtuse d. Exterior
133. What do you call a triangle having three unequal sides?
a. Obtuse b. Oblique **c. Scalene** d. Isosceles
134. How do you call the distance of a point from the y-axis?
a. Polar distance b. Coordinate **c. Abscissa** d. Ordinate
135. This is the measure of central tendency defined as the most frequent score. How do you call this measure of central tendency?
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